

HabitFlow

UI Flow

Fourteen transitions across fifteen screens. Each one names the element that fires it, the destination, the navigation route, the arguments carried, and what the back button does.

NAME	EXAM ROLL	ROLL	PLATFORM	THEME
Shimul Baidya	220419	359	Android · Kotlin	Lifestyle (Habits)



Screen IDs are identical to Document 1 — SCR-04 here is SCR-04 there.

Every flow slide follows the same shape: source screen, trigger, destination screen, and the navigation contract between them.

1 Two screens, one transition

Left is where you are, right is where you land. The numbered badge on the left screen marks the element that fires it.

2 Navigation is a route, not a screen

Compose Navigation is specified as a route string with arguments. The route column is what you type into the NavHost.

3 Not every arrow is navigation

Three transitions change state without leaving the screen — the Coach applying a suggestion, the Coach failing, and the Progress range toggle. Those are marked STATE CHANGE and add nothing to the back stack.

4 Back is specified separately

Some transitions clear the back stack with `popUpTo`, some pop a single entry, some are consumed by a sheet. Never assume the default.

5 Arguments are the contract

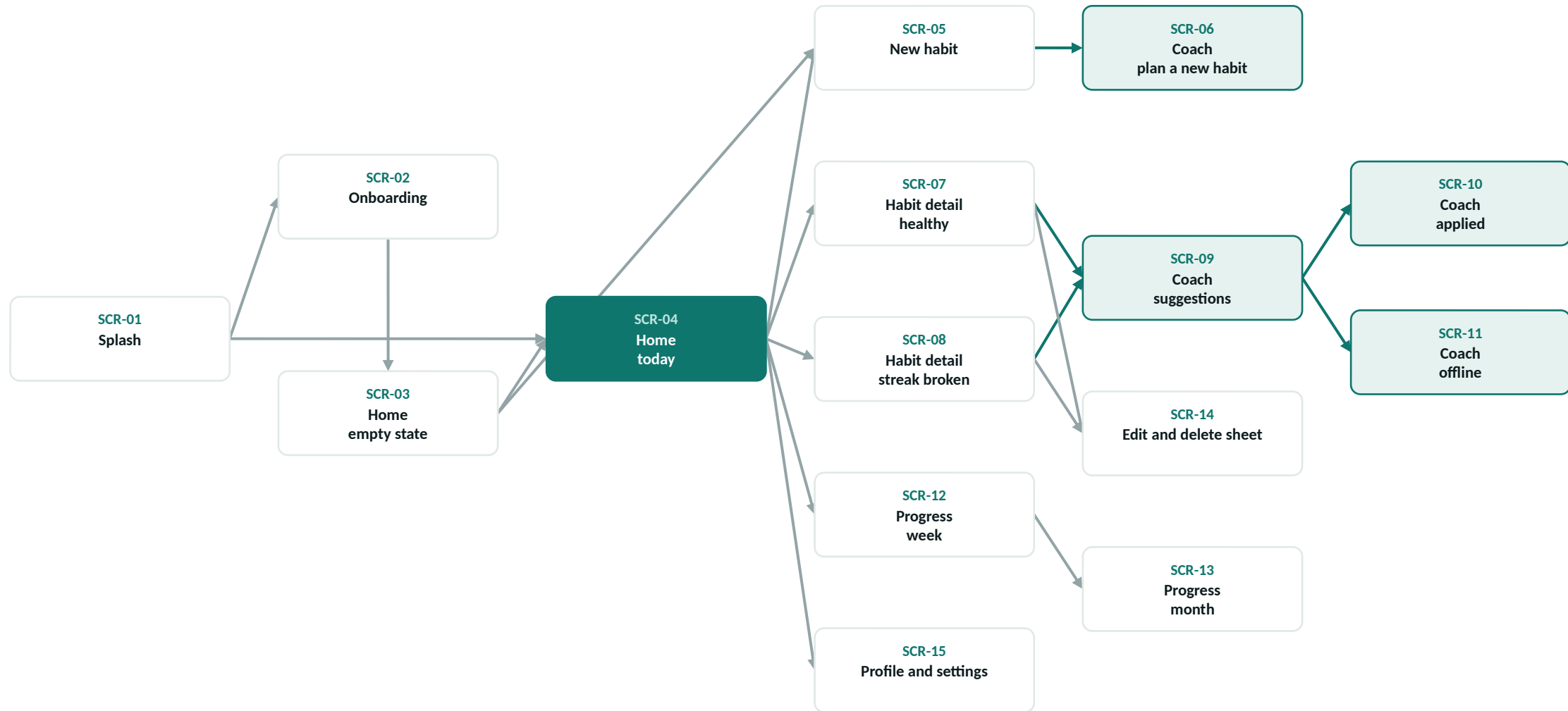
Only the listed arguments cross the boundary. The Coach in particular receives one habit ID and nothing else.

6 The map on slide 3

Shows all fifteen screens and every forward path at once. Back navigation is omitted there for legibility — it is specified per flow instead.

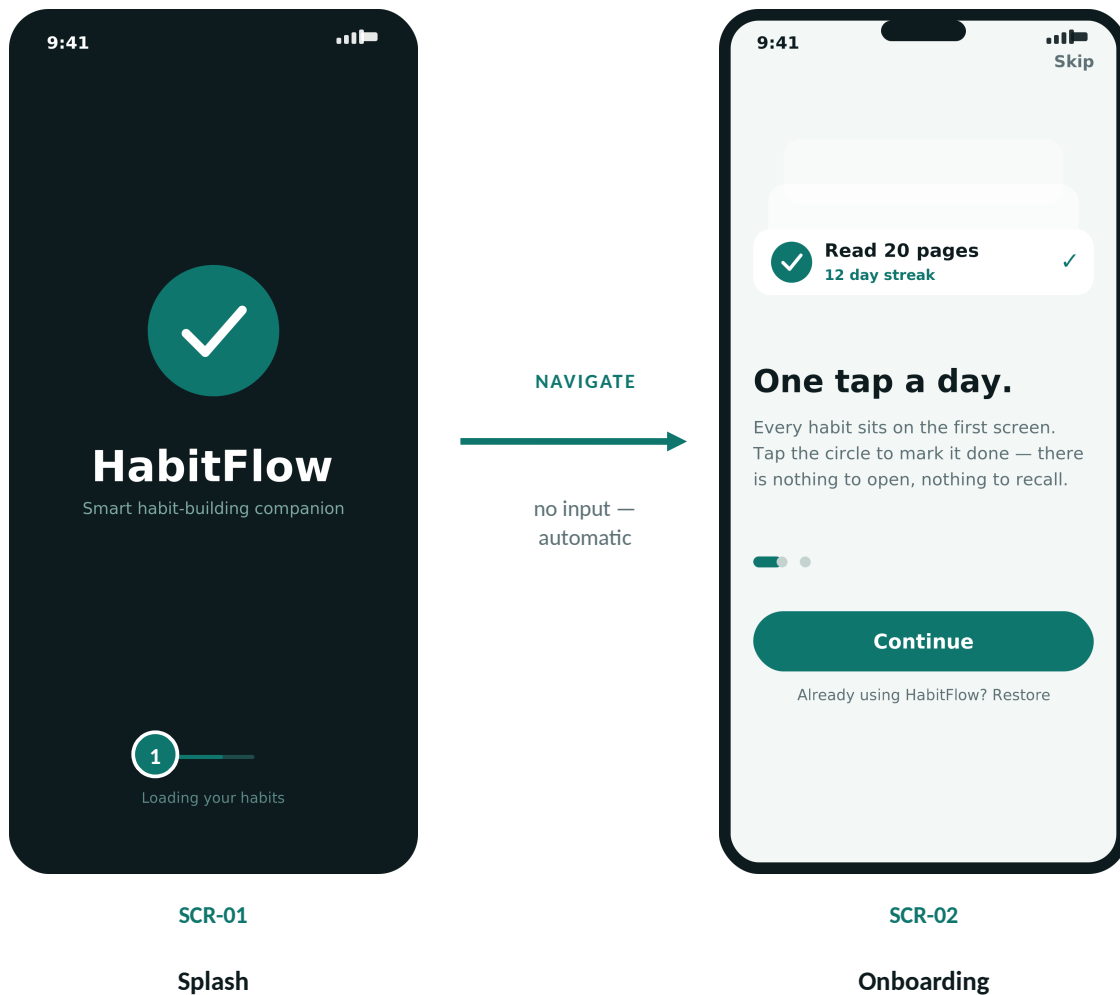
Solid arrow = forward navigation. Dashed = returns with a result. STATE CHANGE = same screen, new state.

All fifteen screens and every forward path. Home is the hub; the Coach is reachable from the two moments that need it — planning a habit and losing one.



Teal = the AI Coach, the one online piece. Grey arrows = local navigation. System back returns to the parent on every path and is specified per flow.

First launch. The only branch in the app decided before a screen is drawn.



TRIGGER

None. LaunchedEffect resolves once Room is open.

WHAT HAPPENS

SettingsDao.onboardingComplete is read. False sends the user to onboarding; true goes straight to Home.

ROUTE

```
navigate("onboarding") {  
  popUpTo("splash") { inclusive = true }  
}
```

ARGUMENTS

None

TRANSITION

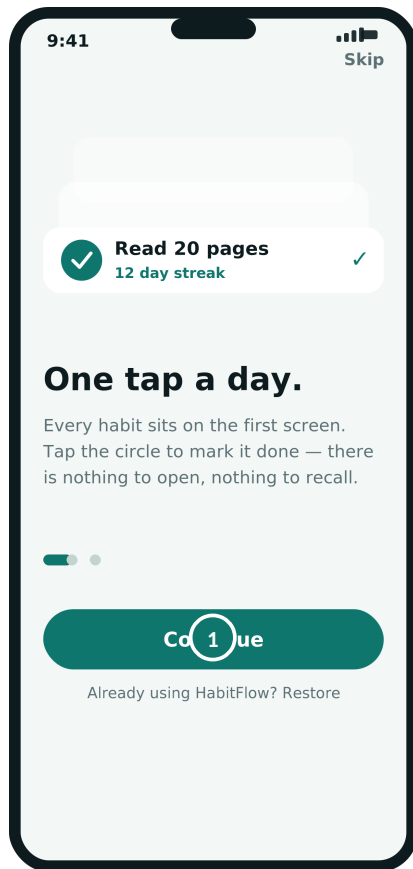
Fade, 200ms. No slide — the splash should not feel like a screen you left.

BACK BEHAVIOUR

Splash is removed from the stack. Back from onboarding exits the app.

The same LaunchedEffect handles the other branch: onboardingComplete = true routes to SCR-04 with the identical popUpTo.

Onboarding finishes into an empty Home, not a populated one.

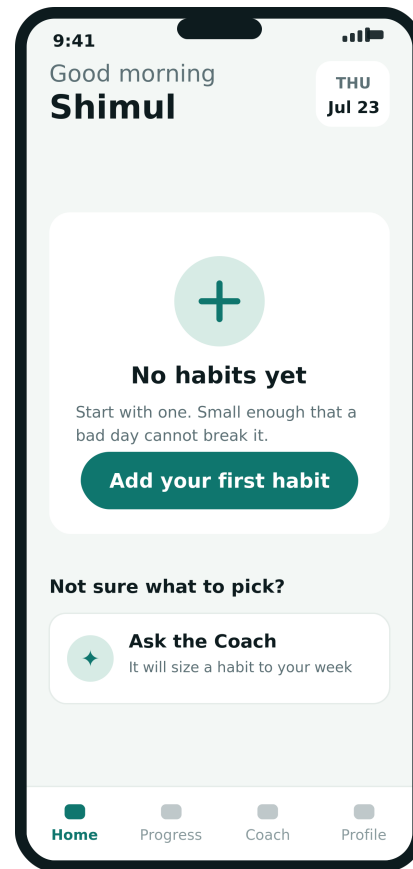


SCR-02

Onboarding

NAVIGATE

tap Continue
on the last pane



SCR-03

Home — empty state

TRIGGER

Continue button (C-08) on pane 3 of 3, or Skip from any pane.

WHAT HAPPENS

onboardingComplete is written to DataStore, then Home opens. No habits exist yet, so Home renders its empty state.

ROUTE

```
navigate("home") {  
  popUpTo("onboarding") { inclusive = true }  
}
```

ARGUMENTS

None

TRANSITION

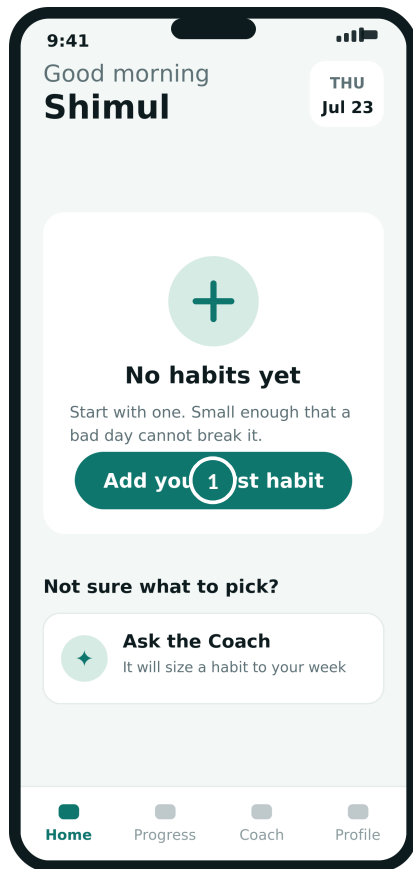
Slide in from the right, 250ms.

BACK BEHAVIOUR

Onboarding is cleared. Back on Home exits the app — it is the start destination.

SCR-03 and SCR-04 are one composable. Which one renders is decided by whether the habit list is empty, not by the route.

The only job of the empty state is to reach this screen.

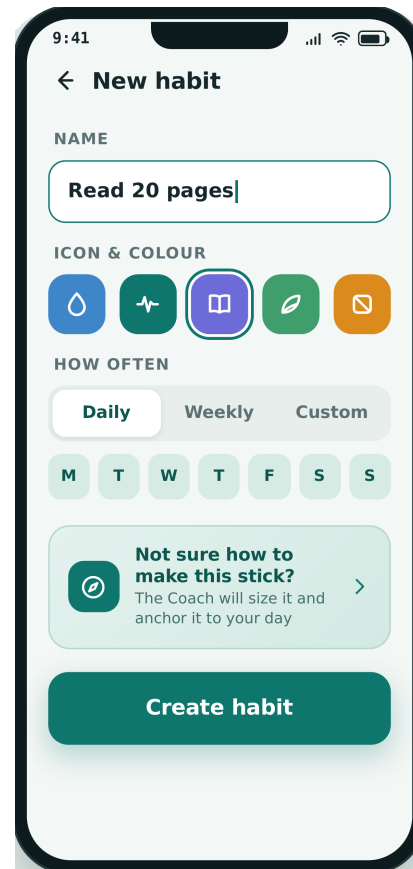


SCR-03

Home — empty state

NAVIGATE

tap Add your first habit



SCR-05

New habit

TRIGGER

Primary button (C-08) inside the empty card.

WHAT HAPPENS

Opens the create form with an empty draft. The same route is used by the FAB on SCR-04.

ROUTE

```
navigate("habit/new")
```

ARGUMENTS

None. A blank HabitDraft is created by the ViewModel.

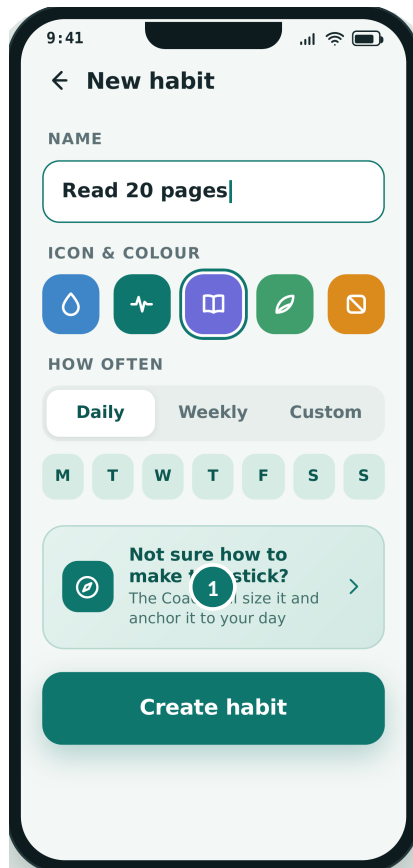
TRANSITION

Slide in from the right, 250ms.

BACK BEHAVIOUR

Pops one entry, returning to Home. If the form is dirty, a confirm dialog intercepts first.

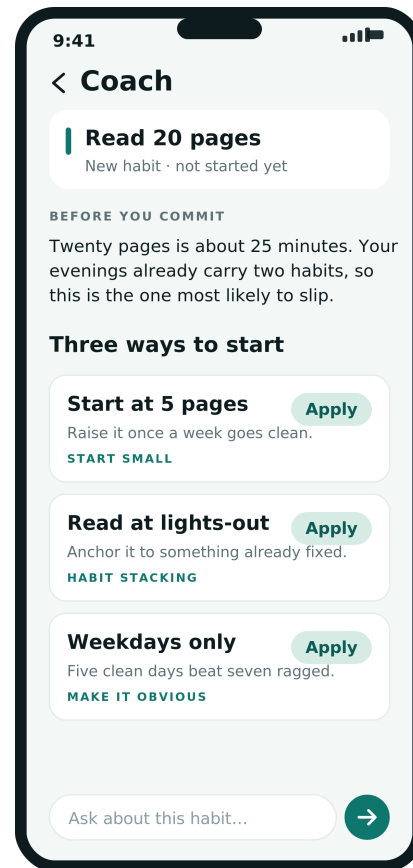
The Coach sizes the habit before it exists — the moment a plan is most useful.



SCR-05

New habit

NAVIGATE

tap the
Coach card

SCR-06

Coach — plan a new habit

TRIGGER

Coach card, positioned directly above Create so it is seen before commitment.

WHAT HAPPENS

The unsaved draft is passed to the Coach, which returns suggested values into the same form.

ROUTE

```
navigate(  
  "coach/plan?name={n}&freq={f}"  
)
```

ARGUMENTS

name, frequency, and the user's current habit count. No habit row exists yet.

TRANSITION

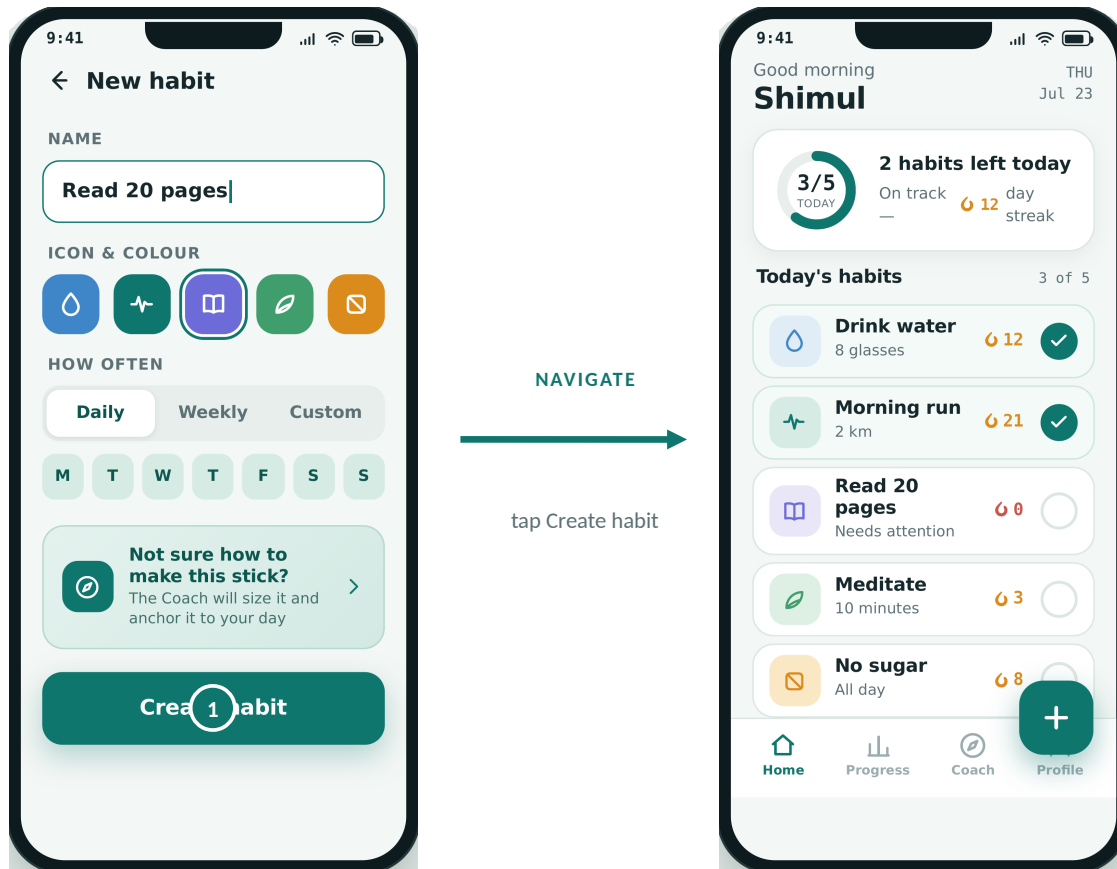
Slide up, 250ms — a modal-feeling detour, not a forward step.

BACK BEHAVIOUR

Pops to SCR-05 with the applied values written into the draft via a `SavedStateHandle` result.

Returning with a result is the reason this is a pop rather than a fresh navigate to SCR-05 — the form state must survive.

Creation. The list is no longer empty, so Home renders its populated state.



SCR-05

New habit

SCR-04

Home — today

TRIGGER

Create button (C-08), enabled only once the name is non-blank.

WHAT HAPPENS

HabitDao.insert writes the row. Home is observing a Flow, so the new habit appears without an explicit refresh.

ROUTE

`popBackStack()`

ARGUMENTS

None. The insert is the message.

TRANSITION

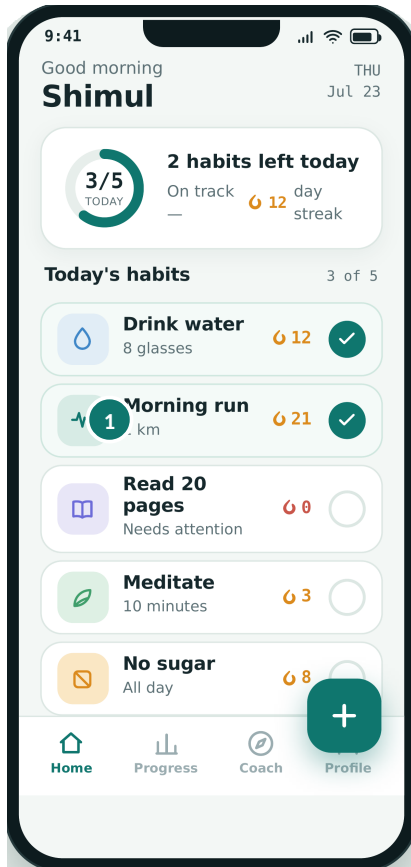
Slide out to the right, 250ms. The new row fades in on Home over 200ms.

BACK BEHAVIOUR

Already at the start destination. Back exits the app.

Crossfade from SCR-03 to SCR-04 runs only on the first insert, when the list goes from empty to non-empty.

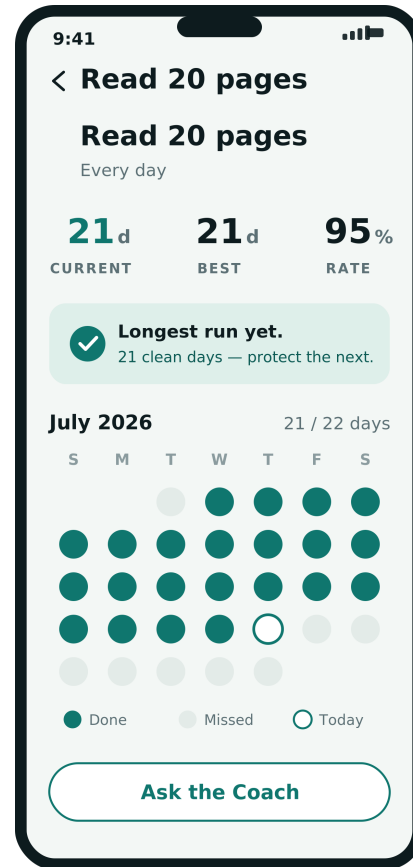
Tapping a habit whose streak is intact opens the reward view.



SCR-04

Home — today

NAVIGATE

tap a habit row
(not the ring)

SCR-07

Habit detail — healthy

TRIGGER

Habit row (C-01). The completion ring inside it is a separate target and does not navigate.

WHAT HAPPENS

Opens the detail screen for that habit. StreakCalculator finds no risk, so the healthy layout renders.

ROUTE

```
navigate("habit/{habitId}")
```

ARGUMENTS

habitId (Long). Everything else is re-queried from Room.

TRANSITION

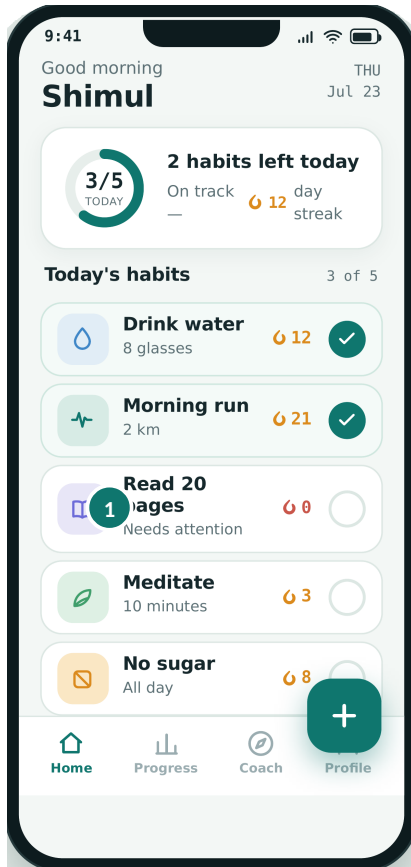
Shared-element on the habit name, 300ms; container transform on the card.

BACK BEHAVIOUR

Pops one entry to Home.

Tapping the ring instead toggles today's completion in place and never leaves SCR-04.

Same tap, same route — a different screen, because the data says so.

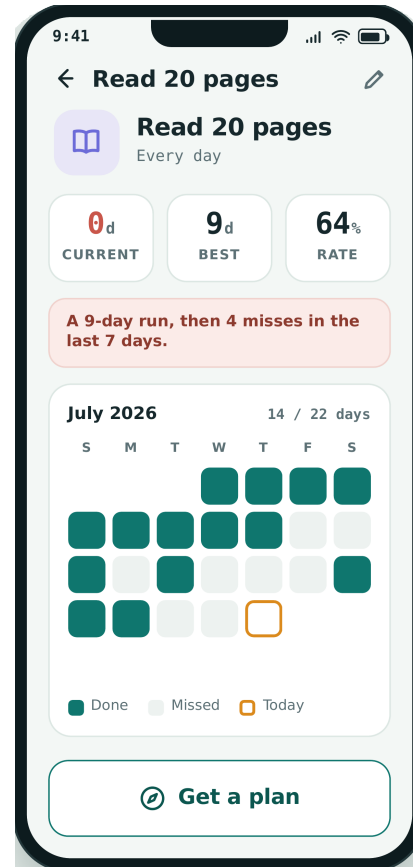


SCR-04

Home — today

NAVIGATE

tap an at-risk
habit row



SCR-08

Habit detail — streak broken

TRIGGER

A habit row sitting under the Needs attention divider.

WHAT HAPPENS

The route is identical to FLOW-06. StreakCalculator finds 2 or more misses in the last 7 scheduled days and the at-risk layout renders instead.

ROUTE

```
navigate("habit/{habitId}")
```

ARGUMENTS

habitId (Long) — the same argument as FLOW-06.

TRANSITION

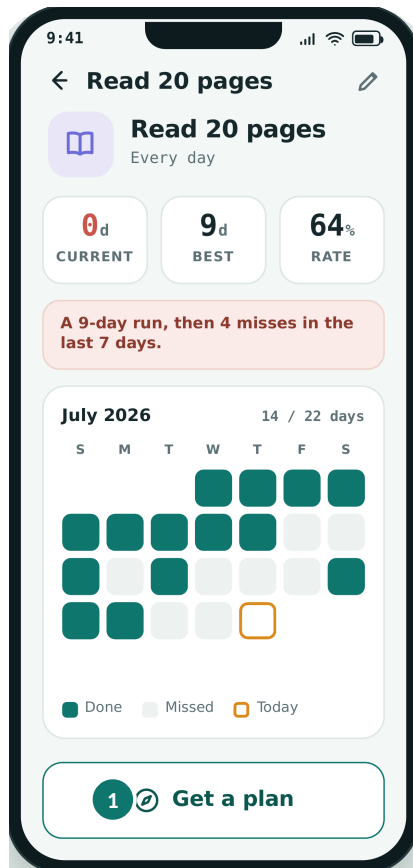
Identical to FLOW-06.

BACK BEHAVIOUR

Pops one entry to Home.

SCR-07 and SCR-08 share one route and one composable. The branch lives in the ViewModel, not the NavHost — build it once.

The transition the project is built around: a broken streak becomes something to do.

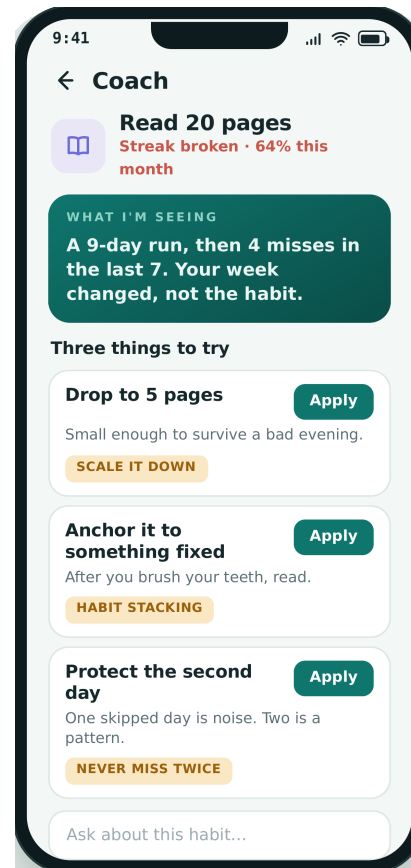


SCR-08

Habit detail — streak broken

NAVIGATE

tap Get a plan



SCR-09

Coach — suggestions

TRIGGER

Get a plan (C-08), filled here and outlined on SCR-07 — the emphasis change is deliberate.

WHAT HAPPENS

StrategyRepository matches the habit's 30-day pattern against roughly 50 local cards, takes the top three, and makes the single network call.

ROUTE

```
navigate("coach/{habitId}")
```

ARGUMENTS

habitId only. The Coach re-derives the pattern summary itself and never receives the habit list.

TRANSITION

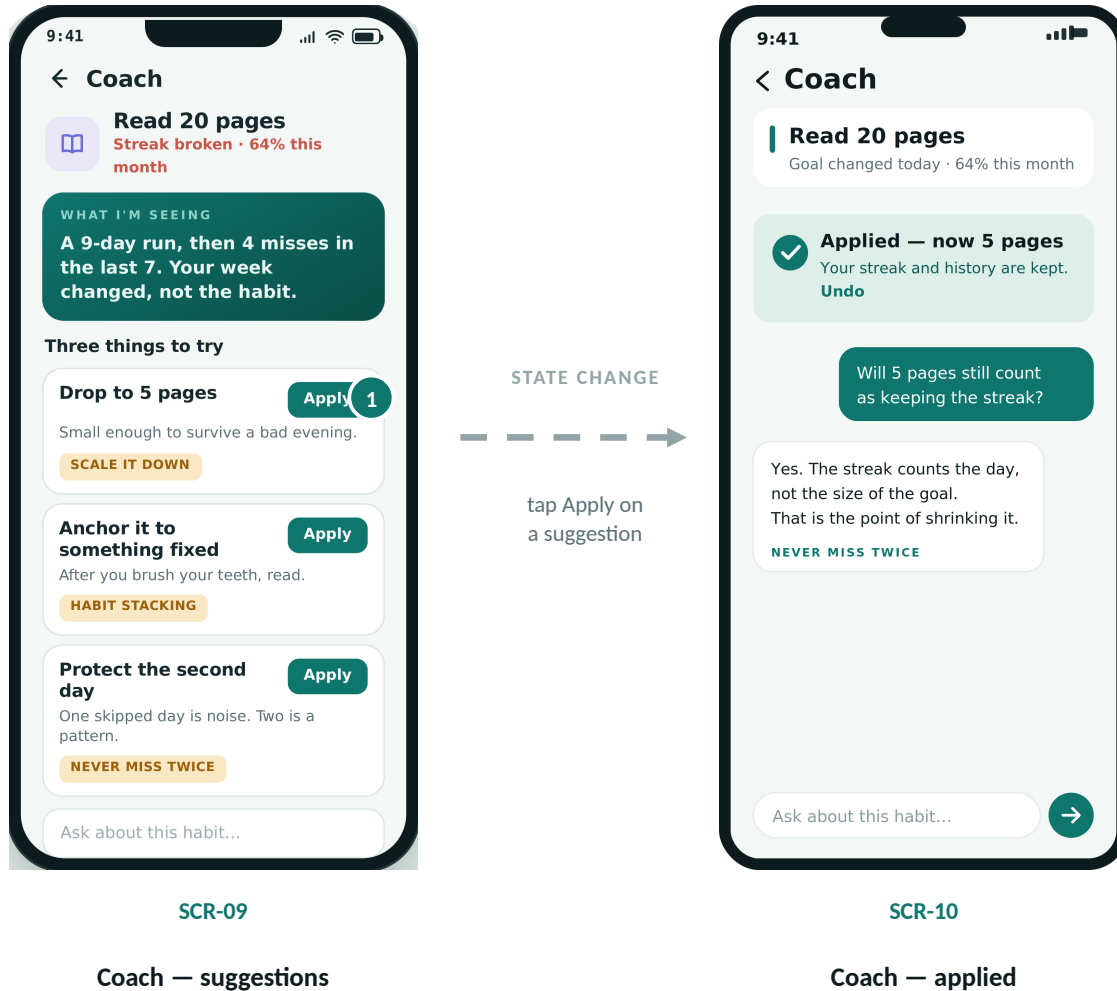
Slide up, 300ms. A three-line shimmer holds the reading area while the call is in flight.

BACK BEHAVIOUR

Pops to the detail screen, which re-queries and reflects any applied change.

Retrieval runs on device before the call. Only the question and one habit's pattern summary leave the phone.

Applying a suggestion writes to Room and never leaves the screen.



TRIGGER

Apply pill (C-07) on one of the three suggestion cards.

WHAT HAPPENS

HabitManager updates the goal. Completions and the streak row are untouched, so shrinking a goal cannot break a streak.

ROUTE

No route. Same composable, new UI state.

ARGUMENTS

suggestionId and the resolved value, held in the Coach ViewModel.

TRANSITION

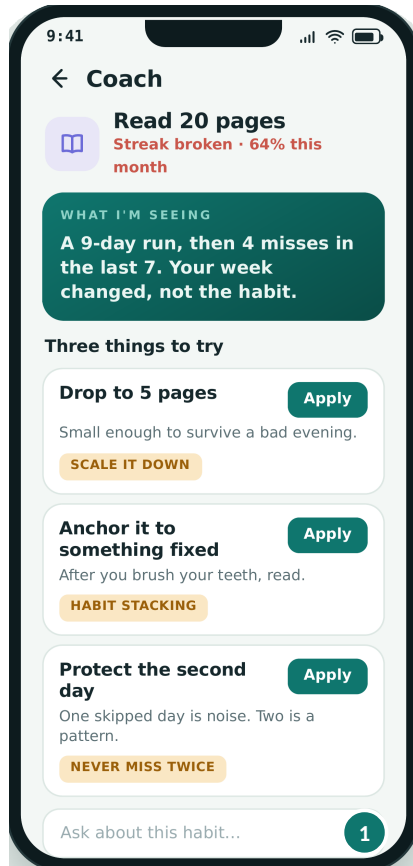
The card collapses into a confirmation row over 250ms; the thread scrolls to it.

BACK BEHAVIOUR

Unchanged — back still pops to the detail screen. Nothing was added to the stack.

A 10-second Undo window is live before the write is committed. Treating this as navigation would break that.

The Coach is the only online piece, so its failure is a designed state.

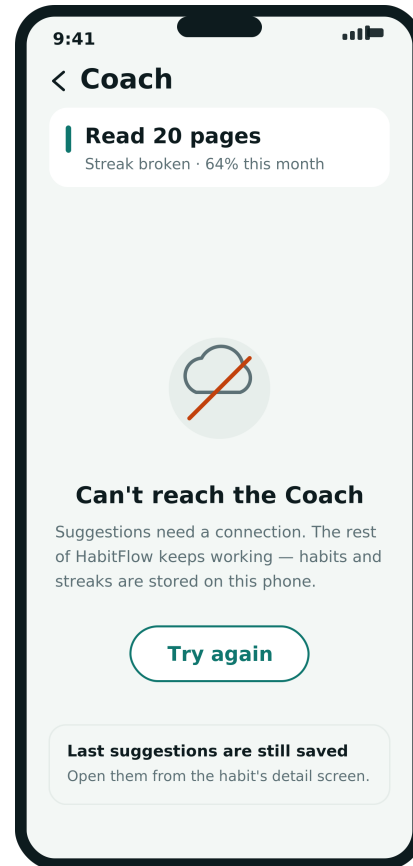


SCR-09

Coach — suggestions

STATE CHANGE

request fails
or no network



SCR-11

Coach — offline

TRIGGER

Not a tap. ConnectivityManager reports no network, or the call times out or is rate limited.

WHAT HAPPENS

The failure state replaces the suggestion list. The habit chip stays, so context survives the error.

ROUTE

No route. Same composable, error state.

ARGUMENTS

None. Cached suggestions are read from CoachSuggestionDao if any exist.

TRANSITION

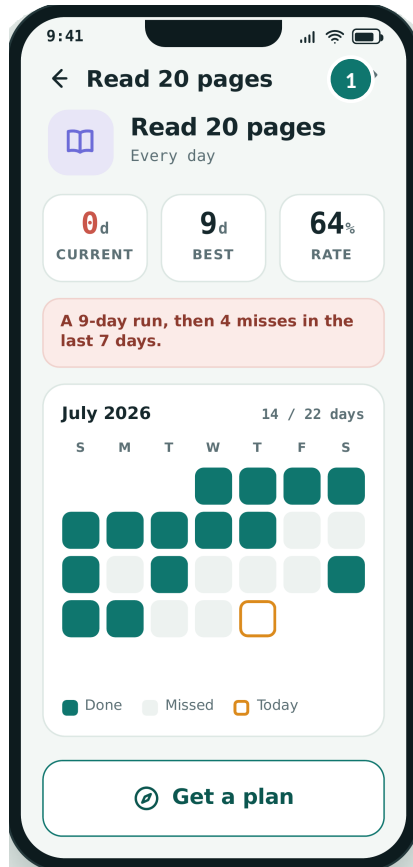
Crossfade, 200ms. No spinner when already offline — it fails fast rather than timing out.

BACK BEHAVIOUR

Unchanged. Retry re-attempts in place with exponential backoff.

Every error variant states that habits and streaks are stored on the device and are unaffected. That sentence is required copy.

Every destructive path on one surface, reached without leaving the habit.

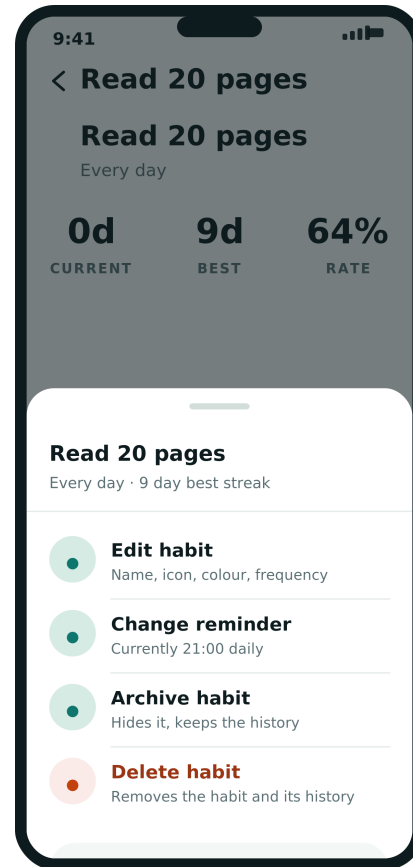


SCR-08

Habit detail — streak broken

NAVIGATE

tap overflow, or
long-press a row



SCR-14

Edit and delete sheet

TRIGGER

App bar overflow on the detail screen, or a long-press on a habit row on SCR-04.

WHAT HAPPENS

A modal bottom sheet opens over the current screen. The habit stays visible above it.

ROUTE

No route. ModalBottomSheet held in the screen's own state.

ARGUMENTS

habitId, already in scope.

TRANSITION

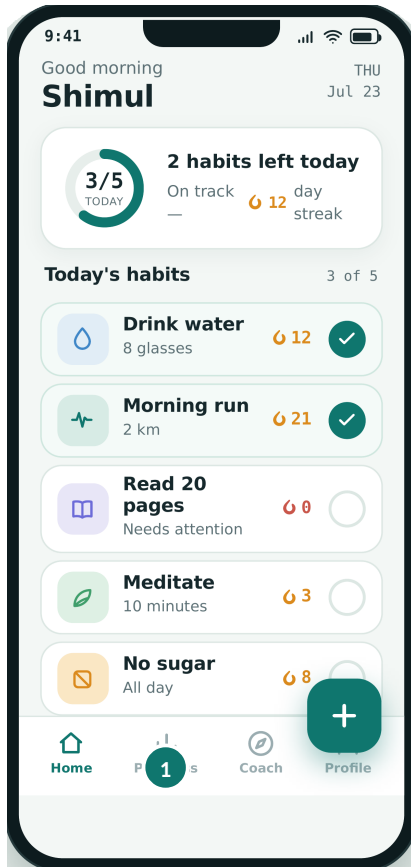
Sheet rises 250ms; scrim fades to 55% over the same duration.

BACK BEHAVIOUR

Back or a tap outside dismisses the sheet and is consumed there — it does not pop the screen.

Delete opens a confirm dialog that names the streak being lost. Archive keeps the history and offers a 5-second undo.

Tab switching. Four roots, one back stack entry between them.

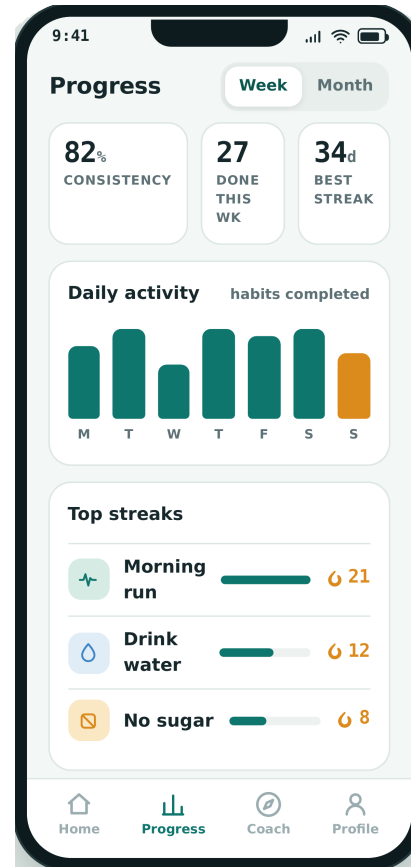


SCR-04

Home — today

NAVIGATE

tap the
Progress tab



SCR-12

Progress — week

TRIGGER

Bottom navigation (C-09).

WHAT HAPPENS

Switches the nav root. Each tab keeps its own scroll position and state.

ROUTE

```

navigate("progress") {
  popUpTo(startDestination) {
    saveState = true
  }
  launchSingleTop = true
  restoreState = true
}

```

TRANSITION

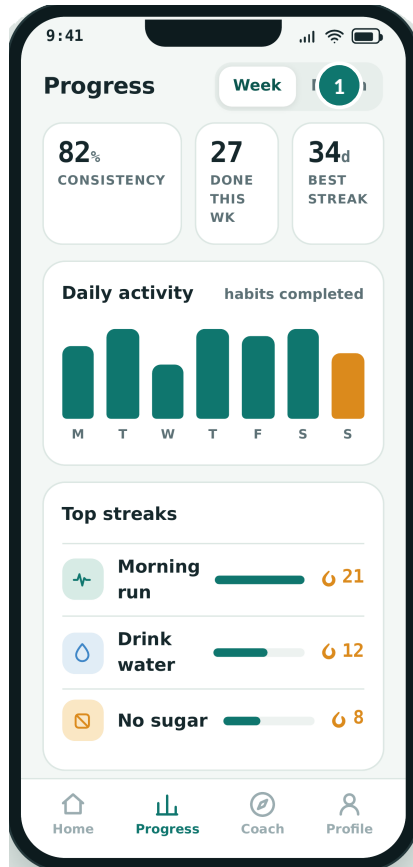
Fade, 150ms. Never a slide — tabs are siblings, not steps.

BACK BEHAVIOUR

Returns to Home from any tab. Tabs never stack on each other.

The Profile tab uses this exact block with route "profile". Specify it once and reuse it for all four tabs.

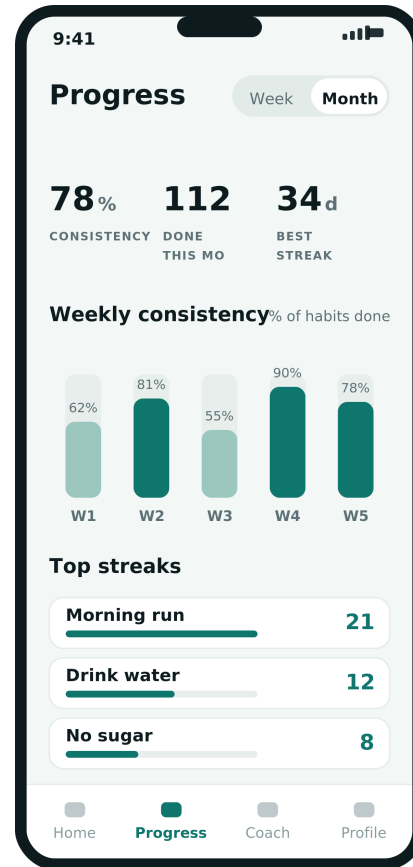
Widening the window is a query, not a destination.



SCR-12

Progress — week

STATE CHANGE
— — — — —
tap Month



SCR-13

Progress — month

TRIGGER

Range toggle (SegmentedButtonRow) in the header.

WHAT HAPPENS

StatsAggregator re-queries over the month instead of the week. Seven daily bars become five weekly ones.

ROUTE

No route. ProgressScreen(range) with a range parameter.

ARGUMENTS

DateTimeRange.MONTH

TRANSITION

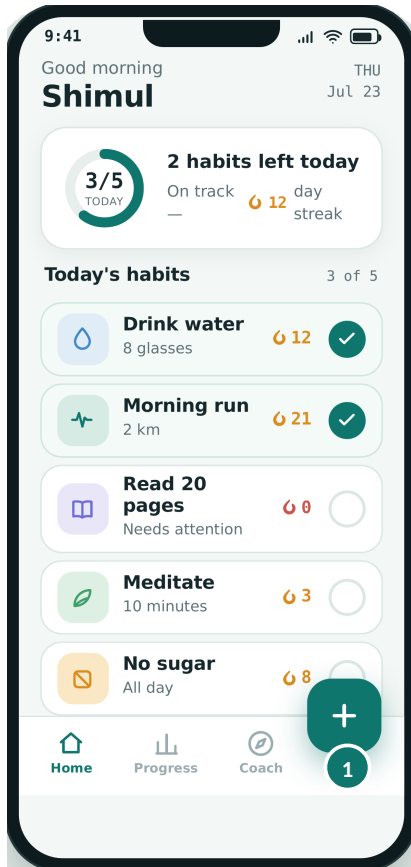
Bars re-animate over 400ms with a 40ms stagger; the stat values count up.

BACK BEHAVIOUR

Unchanged — back leaves Progress entirely and returns to Home.

SCR-12 and SCR-13 are one composable. Two screen IDs exist so the flow is documentable, not because there are two screens to build.

The fourth tab: theme, reminders, and every Coach privacy control.

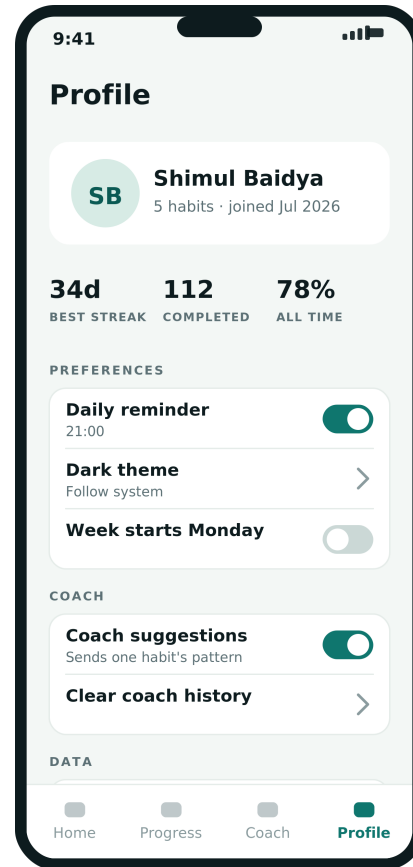


SCR-04

Home — today

NAVIGATE

tap the
Profile tab



SCR-15

Profile and settings

TRIGGER

Bottom navigation (C-09).

WHAT HAPPENS

Opens the settings root. Preference writes go to DataStore immediately — there is no save button.

ROUTE

```
navigate("profile") { ... }  
same block as FLOW-12
```

ARGUMENTS

None.

TRANSITION

Fade, 150ms.

BACK BEHAVIOUR

Returns to Home.

Turning off Coach suggestions here hides every Coach entry point across SCR-03, SCR-05, SCR-07 and SCR-08 — FLOW-04 and FLOW-08 become unreachable by design.